EET202

HINES & TRANSFORMERS

MODULE: 4

cture:3 – Phasor Diagram

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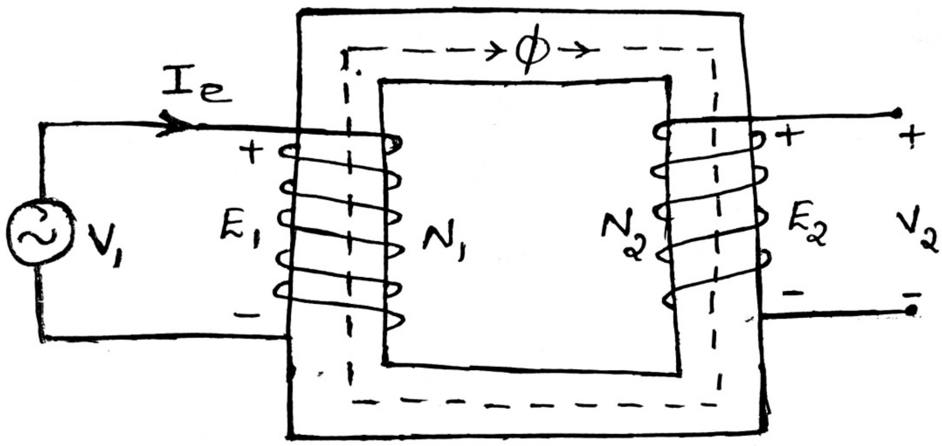
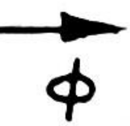
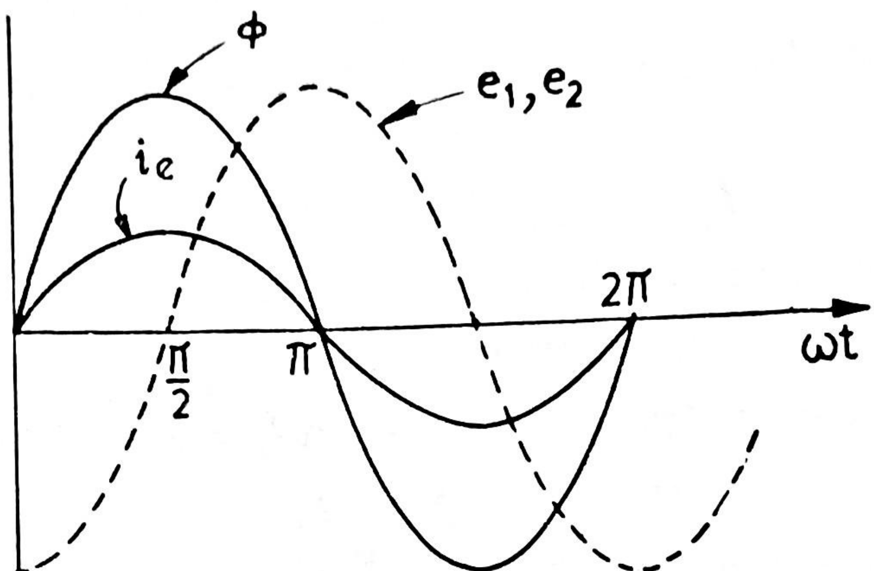
Syllabus

ers –constructional details, principle of operation, EMF equation,
convention, magnetising current, transformation ratio, phasor
no load and on load, equivalent circuit, percentage and per unit
egulation. Transformer losses and efficiency, condition for
/A rating. Testing of transformers– polarity test, open circuit test,
npner's test – separation of losses, all day efficiency.Parallel
se transformers– numerical problems

Transformer on No-Load

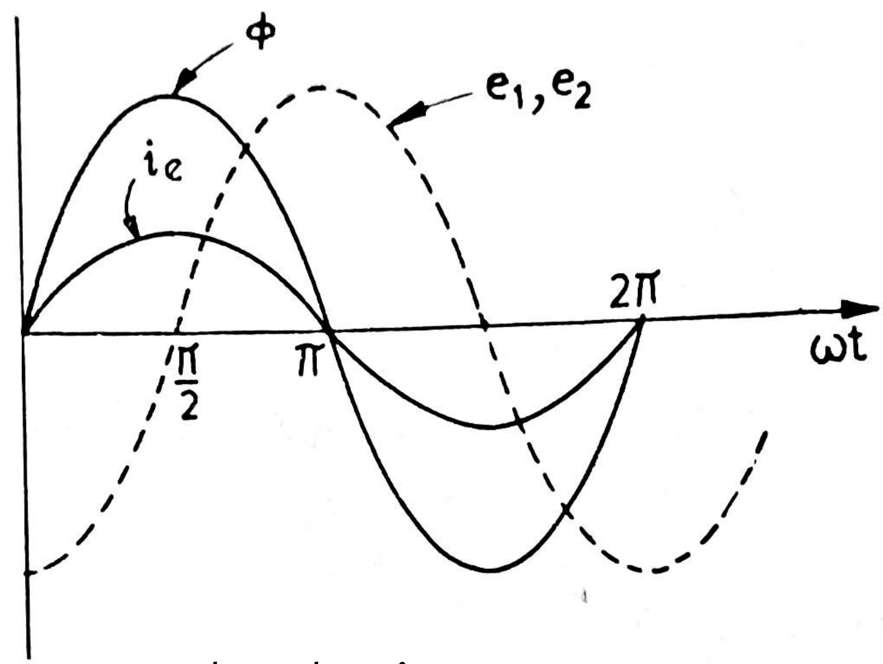






$$= N_1 \phi_m 2\pi f \sin(\omega t - 90)$$

Transformer on No-Load

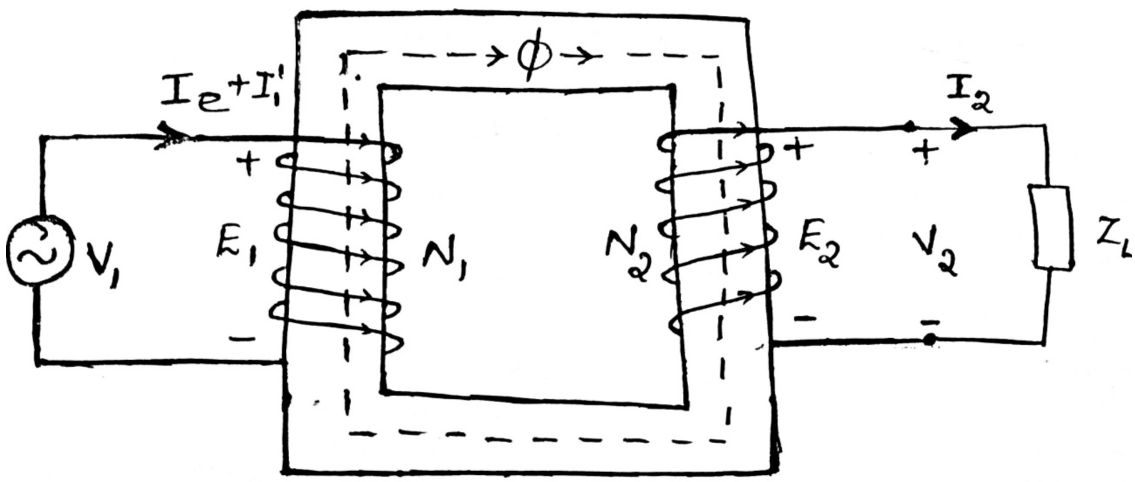


$$\phi = \phi_m \sin \omega t$$

$$e_1 = N_1 \phi_m 2\pi f \sin(\omega t - 90)$$

$$e_2 = N_2 \phi_m 2\pi f \sin(\omega t - 90)$$

l Transformer on Load



$$\phi = \phi_m \sin \omega t$$

$$e_1 = N_1 \phi_m 2\pi f \sin(\omega t - 90)$$

$$e_2 = N_2 \phi_m 2\pi f \sin(\omega t - 90)$$

l Transformer on Load

er is loaded, a current I_2 flows through the ng.

roduce a flux, and it will oppose the main flux ϕ.

flux, will cause reduction in E_1.

nsformer, $V_1 = -E_1$

ant, E_1 can not be varied.

current I_1' will be drawn from the supply to reduction in main flux.

current I_1 will be equal to the vector sum of I_1'

$$\overrightarrow{I_e} + \overrightarrow{I_1'}$$

l Transformer on No-Load

nsformer, there will be following imperfections:

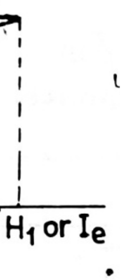
sformer core loss: Current I_e will not be in phase
lux lags behind the current.

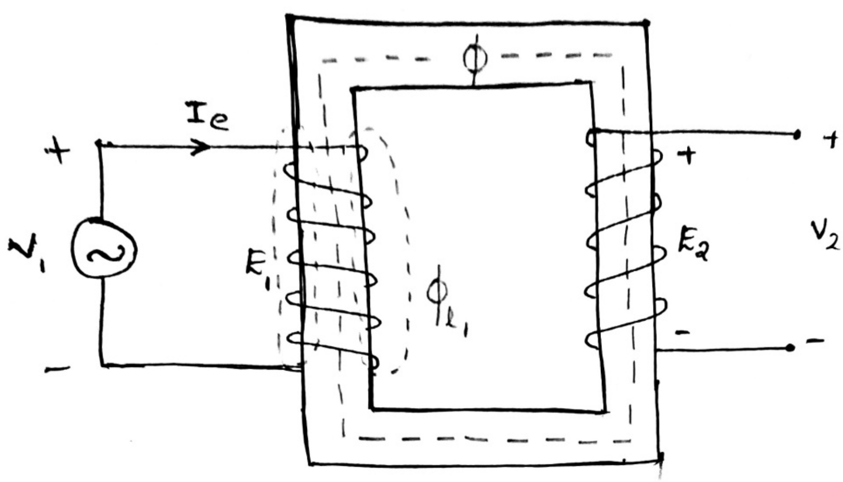
ding resistance: In the primary side there will be
o across this winding resistance.

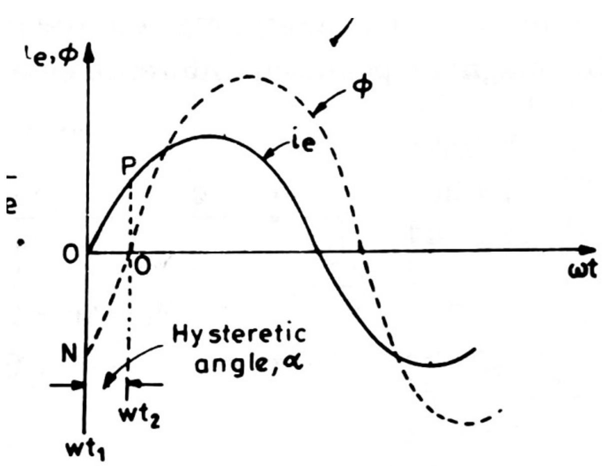
kage flux: This leakage flux will induce an emf in
vinding and according to Lenz’s law, it will oppose
is can be represented as a reactance acting along

$$+\overline{I_e} r_1 + j\overline{I_e} x_1$$

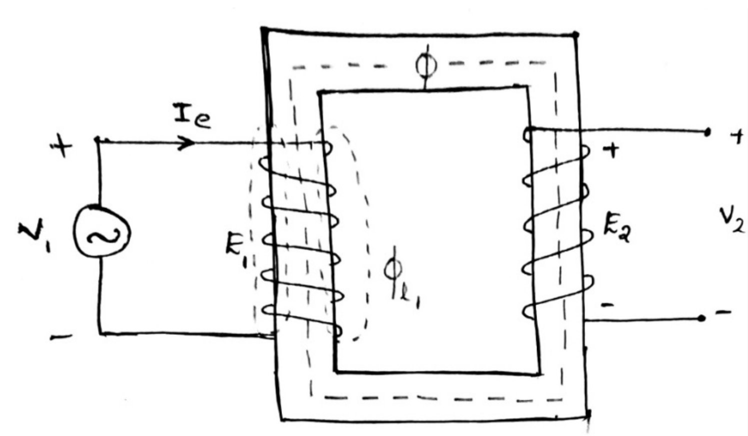
l Transformer on No-Load







l Transformer on No-Load



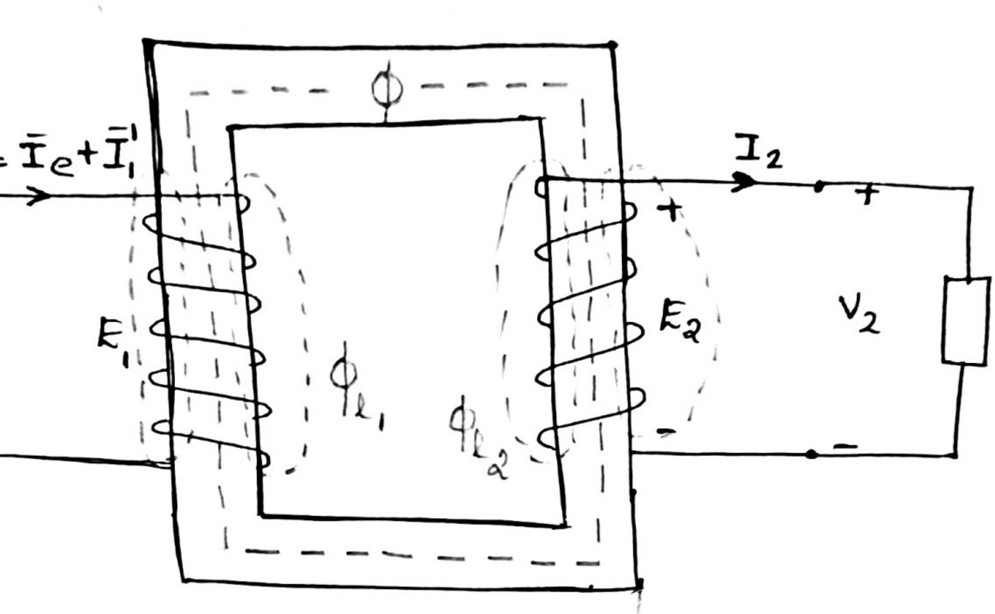
ϕ

I_e is called as **no-load current** or **exciting current**.

It can be resolved into two components.

- Along x-axis: $\mathbf{I_m}$ **– Magnetizing component**
- Along y-axis: $\mathbf{I_c}$ **– Core loss component**

cal Transformer on Load



cal Transformer on Load

er is loaded, a current I_2 flows through the
ng.

winding resistance and leakage reactance at
Hence, $\overline{E_2} = \overline{V_2} + \overline{I_2} r_2 + j\overline{I_2} x_2$

urrent fill produce a flux, and it will oppose the

is demagnetizing effect, an extra current(I_1’) is
supply.

$- \overline{I_1}'$

ing resistance and leakage reactance at primary

$+ \overline{I_1} r_1 + j\overline{I_1} x_1$

cal Transformer on Load

am

erence.

$_e$, which leads the flux by some angle α.

components. I_m along x axis and I_c along y axis.

by 90°. Reverse E_1 to get $-E_1$.

rticular direction.

to the nature of load.

o get E_2.

'.

et I_1.

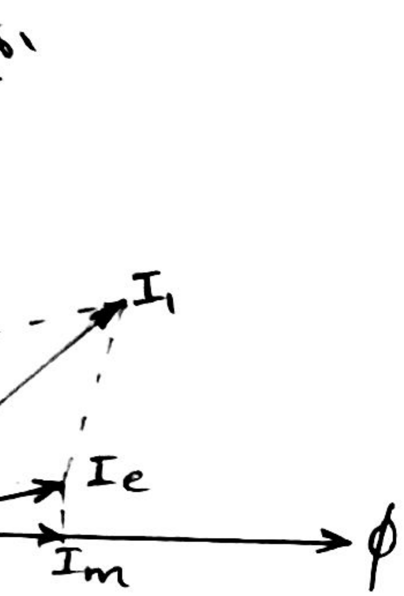
o $-E_1$ to get V_1.

and I_1 can be marked as θ_1 and ($\cos\theta_1$) will be the

tor.

- ✔ For Resistive load (upf):
  I_2 and V_2 are in phase
- ✔ For RL load (lagging pf):
  I_2 lags V_2
- ✔ For RC load (leading pf):
  I_2 leads V_2

Transformer on Unity PF Load



ransformer on Lagging PF Load

ransformer on Leading PF Load